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In [1]: # Name - Rohan Pise  
# Rollno. - 01
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In [3]: import numpy as np  
X = np.array([[1, 1, 1, -1], [-1, -1, 1, 1]])  
Y = np.array([[1, -1], [-1, 1]])  
W = np.dot(Y.T, X)  
  
def bam(x):  
    return np.sign(np.dot(W, x))  
  
x_test = np.array([1, -1, -1, -1])  
y_test = bam(x_test)  
  
print("Input x:", x_test)  
print("Output y:", y_test)
```

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Input x: [ 1 -1 -1 -1]  
Output y: [ 1 -1]
```

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In [ ]:
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